Finite Groups and Their Representations — Exercise Sheet 3

Sylow theory and small groups

Variant: Questions only

Exercise 1 (Order thirty). Let $|G| = 30$. List the possibilities for n_2, n_3, n_5 allowed by Sylow's theorems. Prove that G has a normal Sylow 3-subgroup or a normal Sylow 5-subgroup. Deduce that G is not simple.

Exercise 2 (Order fifty-six). Let $|G| = 56$. Prove that G is not simple. In the case $n_7 > 1$, prove that there is exactly one Sylow 2-subgroup.

Exercise 3 (A subgroup that is not Sylow). In S_4, let $H = \langle (1\,2) \rangle$. Find all conjugates of H and its normaliser. Explain why their number does not contradict Sylow's congruence modulo 2.

Exercise 4 (The Frattini argument). Let G be finite, let $N \trianglelefteq G$, and let P be a Sylow p-subgroup of N. Prove $G = N N_G(P)$. Apply the result to $G = S_5$, $N = A_5$ and $P = \langle (1\,2\,3) \rangle$. Find $N_G(P)$ and the order of $N \cap N_G(P)$.

Exercise 5 (The non-abelian group of order pq). Let $p < q$ be odd primes with $p \mid q - 1$. You may use the elementary fact that a non-zero polynomial of degree d over a field has at most d roots. Use Sheet 2's identification $\mathrm{Aut}(C_q) \cong \mathbb{F}_q^{\times}$ and its external semidirect-product construction.

(i) Prove that a finite subgroup A of the multiplicative group of a field is cyclic. First let m be the least common multiple of its element orders. For each prime dividing m, choose an element whose order attains the highest power of that prime in m. Take a suitable power of each chosen element, and combine the resulting elements of coprime orders.

(ii) Deduce that $\mathrm{Aut}(C_q)$ is cyclic of order $q - 1$. Construct a non-abelian group $C_q \rtimes C_p$, and prove its uniqueness up to isomorphism among non-abelian groups of order pq.

(iii) For $p = 3, q = 7$, write an explicit multiplication on $\{(a, b) : a \pmod 7,\ b \pmod 3\}$ and find its centre.

Exercise 6 (Groups of order eight). Classify the groups of order 8 up to isomorphism. Write C_2^3 for the direct product of three copies of C_2.

(i) In the abelian case, distinguish an element of order 8, an element of order 4, and the case where every non-identity element has order 2.

(ii) In the non-abelian case, prove that an element r of order 4 exists. For $s \notin \langle r \rangle$, show $srs^{-1} = r^{-1}$ and $s^2 = e$ or r^2. Identify both multiplication rules, comparing with the Q_8 matrix model generated by a and b on Sheet 2.

(iii) Distinguish the resulting groups using commutativity and the number of non-identity elements of order 2.

Exercise 7 ⋆. Classify the groups of order 12, using these steps.

(i) If $n_3 = 4$, use Lecture 3 to obtain a normal subgroup P of order 4. Show that $G = P \rtimes C_3$ and that the only possibility is A_4. For this purpose prove $\mathrm{Aut}(V_4) \cong S_3$ by permuting its three non-identity elements.

(ii) If $n_3 = 1$, write $G = C_3 \rtimes P$ with $|P| = 4$. For $P \cong C_4$ and $P \cong V_4$, classify homomorphisms $P \to \mathrm{Aut}(C_3) \cong C_2$ up to changing generators.

(iii) Distinguish the five resulting groups.